

Henry J. Nelson

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Education

PhD	University of Minnesota , Computer Science Thesis: In-Situ Maize Reconstruction and Analysis for Precision Agriculture	Minneapolis, MN Aug 2017 – Nov 2025
BA	Grinnell College , Physics	Grinnell, IA Aug 2013 – May 2017

Industry Experience

Deere & Company (John Deere) , Staff Robotics Engineer Development, training, evaluation, and deployment of machine learning and computer vision algorithms for weed detection, localization, and species classification. Supporting both cloud infrastructure and edge devices. Lead the machine learning efforts behind the SMARTSCRIPT weeds product.	Saint Paul, MN May 2025 – present
Sentera , Machine Learning Engineer Deep learning infrastructure and model development, training, and deployment. Algorithm development for automatic analysis and interpretation of drone imagery for applications in precision agriculture. Using both traditional computer vision (3D geometry and image processing) and machine learning approaches. Startup acquired by John Deere in May 2025.	Minneapolis, MN May 2019 – May 2025

Teaching Experience

University of Minnesota, Department of Computer Science , Teaching Assistant Preparing and giving weekly lectures, managing other TAs, grading, and holding regular office hours for both undergraduate and graduate level courses. Courses: Automata and Formal Languages; Computer Vision; Artificial Intelligence; and Algorithms and Data Structures.	Minneapolis, MN Aug 2017 – Dec 2019
Grinnell College, Department of Computer Science , Teaching Assistant Preparing and giving weekly lectures. Providing supplemental resources and individualized instruction. Courses: Automata, Formal Languages, and Computational Complexity.	Grinnell, IA Aug 2016 – May 2017
Grinnell College, Department of Physics , Teaching Assistant Instruction of introductory, intermediate, and upper level students in course content, lab preparation, experiment execution, and data analysis in both classroom and laboratory settings. Courses: Mechanics; and Electrostatics.	Grinnell, IA Aug 2015 – May 2016

Skills

Programming Languages: Python, MATLAB, C/C++, Julia
Libraries: PyTorch, Tensorflow, NumPy, Numba, OpenCV, Open3D
Software Development and Deployment Tools: Git, Docker, AWS

Awards

Best Agricultural-Robotics Paper Finalist IROS 2024	Oct 2024
H. George Apostle Prize in Physics Grinnell College, Department of Physics	May 2017
Phi-Beta-Kappa Grinnell College	May 2017

Patents

Precision agriculture using pose georeferenced analytics (US12310274B1) Andrew Muehlfeld, Henry J. Nelson , Ryan Nelson, Eric Taipale, Nicholas Witthoeft	May 2025
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Publications

Efficient 3D Reconstruction in Noisy Agricultural Environments: A Bayesian Optimization Perspective for View Planning Athanasios Bacharis, Konstantinos D. Polyzos, Henry J. Nelson , Georgios B. Giannakis, Nikolaos Papanikolopoulos 10.1109/LRA.2025.3625501 ↗ (IEEE Robotics and Automation Letters)	Oct 2025
Ground-Density Clustering for Approximate Agricultural Field Segmentation Henry J. Nelson , Nikolaos Papanikolopoulos 10.1109/IROS58592.2024.10802415 ↗ (IEEE/RSJ IROS 2024)	Oct 2024
Robust Plant Localization and Phenotyping in Dense 3D Point Clouds for Precision Agriculture Henry J. Nelson , Christopher E. Smith, Athanasios Bacharis, Nikolaos Papanikolopoulos 10.1109/ICRA48891.2023.10161078 ↗ (IEEE ICRA 2023)	May 2023
View Planning Using Discrete Optimization for 3D Reconstruction of Row Crops Athanasios Bacharis, Henry J. Nelson , Nikolaos Papanikolopoulos 10.1109/IROS47612.2022.9981209 ↗ (IEEE/RSJ IROS 2022)	Oct 2022
A Methodology for the Detection of Nitrogen Deficiency in Corn Fields Using High-Resolution RGB Imagery Dimitris Zermas, Henry J. Nelson , Panagiotis Stanitsas, Vassilios Morellas, David J. Mulla, Nikolaos Papanikolopoulos 10.1109/TASE.2020.3022868 ↗ (IEEE Transactions on Automation Science and Engineering)	Oct 2021
Learning Continuous Object Representations from Point Cloud Data Henry J. Nelson , Nikolaos Papanikolopoulos 10.1109/IROS45743.2020.9341765 ↗ (IEEE/RSJ IROS 2020)	Feb 2021
Weed Detection and Classification in High Altitude Aerial Images for Robot-Based Precision Agriculture Karthik Buddha, Henry J. Nelson , Dimitris Zermas, Nikolaos Papanikolopoulos 10.1109/MED.2019.8798582 ↗ (IEEE MED 2019)	July 2019